

Hands-on Lab: Run Malware Scans Using rkhunter

Estimated time needed: 20 minutes



In this project, you will install rootkit hunter (rkhunter) in Linux and run a malware scan.

Objectives

After completing this lab, you will be able to:

- Install rkhunter on a Linux system
- Run a malware scan to check for rootkits and malware

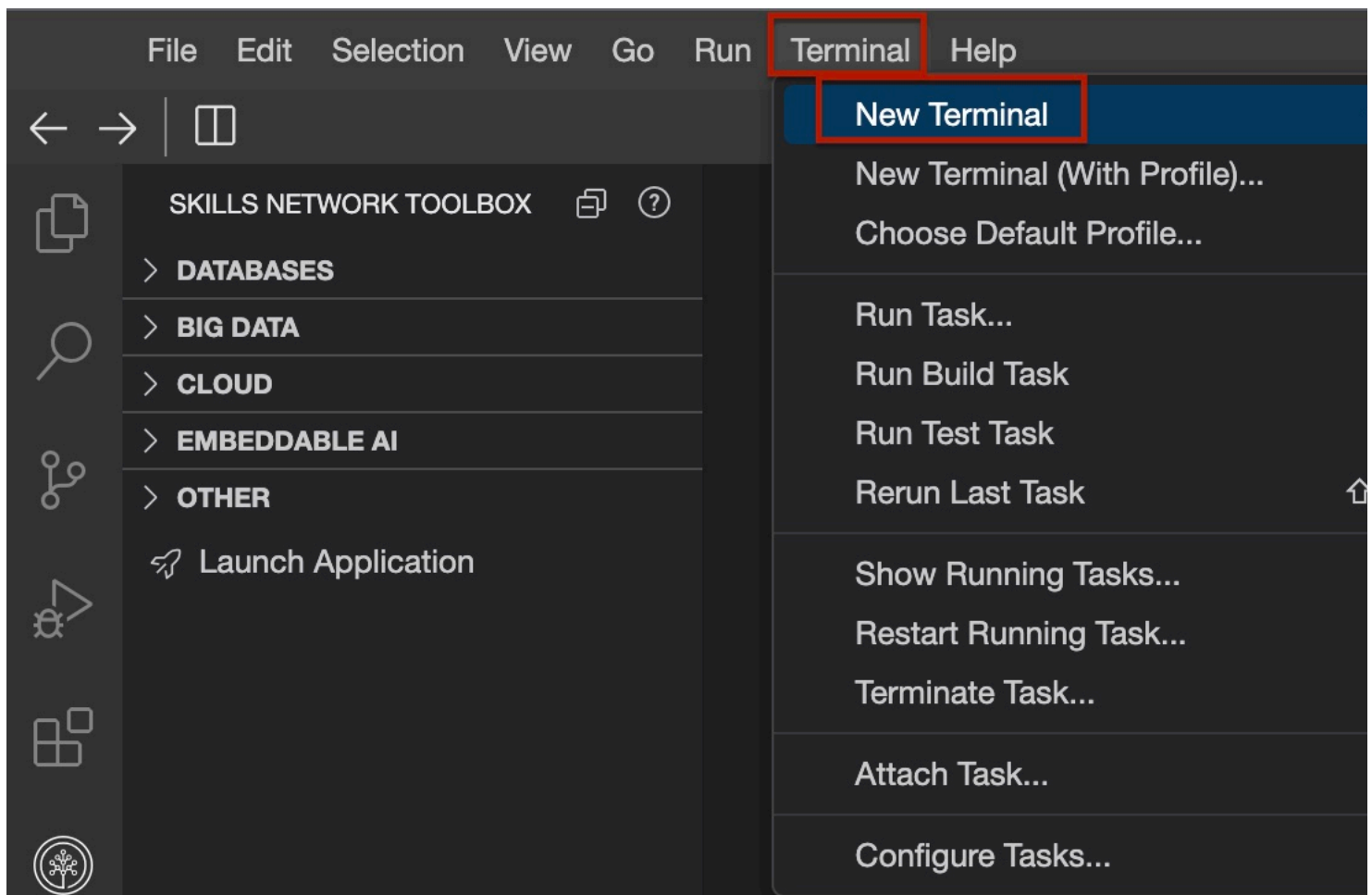
Prerequisites

This lab requires Docker.

Install Rkhunter

Rkhunter, short for rootkit hunter, is a Linux/Unix-based utility that detects rootkits, backdoors, and potential local exploits. In addition to running on-demand scans, it can alert administrators to any unusual activity within the server's file system. Let's see how to install rkhunter.

1. Open a new terminal.



2. In the new terminal, run the following command.

```
sudo apt-get update
```

This will take a few seconds to complete the installation.

3. Run the following command to install rkhunter.

```
sudo apt-get install rkhunter
```

1. While installing, when prompted with the confirmation query “Do you want to continue? [Y/n],”choose option “y” and press enter.

2. Subsequently, you will encounter a prompt to choose the preferred mail server configuration type. The options include:

- 1.No configuration
- 2.Internet Site
- 3.Internet with smarthost
- 4.Satellite system
- 5.Local only

For the general mail configuration type, select option 1 and press enter.

4. Once installed, you will need to update the file properties database. This action informs the scanner about the state of system files to prevent false alarms. The –**propupd** command generates a clean **rkhunter.dat** file that is used from this point forward as a baseline for further checks. Run the following command to update the database.

```
sudo rkhunter --propupd
```

Run rkhunter scan

1. Run the following command to start the scan. The first scan will be to check the system commands. As the scan runs, you will see the progress on the screen.

```
sudo rkhunter --checkall
```

2. Once the system command scan is complete, select **Enter** to start the scan, which will look for rootkits. As the scan progresses, you will see that the scanner is looking for registry keys, log files, and directories associated with known threats.
3. Once the rootkit scan is complete, select **Enter**. Rkhunter will now perform additional rootkit, malware, and Linux-specific checks.
4. Once the additional checks are complete, select **Enter** to perform network and local host checks.
5. Once the network and local host checks are complete, select **Enter** to see a scan summary.

Review scan log

1. To see a full view of all scan results, run the following command.

```
sudo cat /var/log/rkhunter.log
```

2. To see a condensed view of all scans that result in a warning, run the following command.

```
sudo cat /var/log/rkhunter.log | grep -i warning
```

Note that many of the warnings you see in this lab are a result of the controlled environment and would look different in a live setting. However, you can see how the provided warnings could serve as a starting point for better securing your system.

Conclusion

Congratulations! In this lab, you learned to run a scan using rkhunter and review the resulting log file.

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